The Paleolithic Age

Early Humans, Tools, and the Old Stone Age



A Survival Guide to the Ancient World

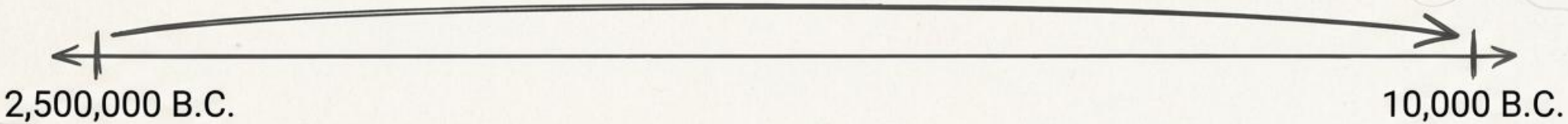
TEACHER NOTES:

Welcome students.
Ask: If you had to live in the wild with only tools you made yourself, could you do it?
Explain this notebook will reveal the secrets of early human survival.

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What is the Paleolithic Age?

Key Vocabulary: **Paleolithic** = Greek word for **old stone**.
Also known as: **The Old Stone Age**.



The basic feature of the human species is the ability to make tools.

TEACHER NOTES:
Break down the word: Paleo = old, lithic = stone.
Emphasize how incredibly long this time period is.
The main takeaway: making tools makes us human.

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Finding Food: Hunting and Gathering

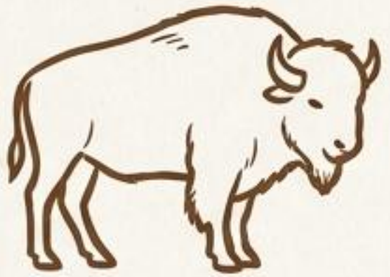
Early humans had a very close relationship with nature. They survived by taking what nature provided.

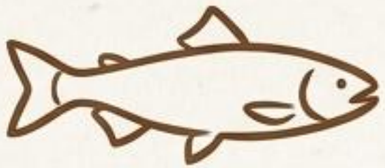
Gathering



Wild nuts, berries, fruits, wild grains, green plants.

Hunting





Buffalo, horses, bison, reindeer.

In coastal areas: fish and shellfish.

TEACHER NOTES:
Explain they had no farms or stores.
Ask: Which do you think was easier: hunting large buffalo or gathering berries? Why?

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The Paleolithic Way of Life: First Tools

Simple stone tools helped early humans survive.



1. Flint

Using hard stones. One stone chips another to create a sharp edge.

2. Hand Axes

Pointed tools with cuttin edges to cut plants and clean animal hides.

3. Spears

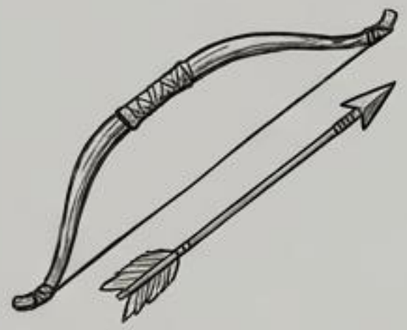
Tying a stone point to a wooden pole to kill large animals.

TEACHER NOTES:
Walk through the invention process. Define “flint” and “hides” (animal skins). Show how combining wood and stone created powerful new technology.

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Later Inventions: Better Tools

Over thousands of years, tools became smarter and hunting became easier.



Bow and Arrow

Made hunting much easier from a safe distance.



Harpoons

Made of bone. Used for catching large fish.



Fishhooks

Made of bone. Used to catch smaller fish.

Bone Needles

Used to make nets, baskets, and sew animal hides for clothing.

TEACHER NOTES:
Ask: Why is bone better than stone for needles? (It can be carved into small, sharp, curved shapes). Sewing clothes was a massive leap for survival.

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Nomadic Life and Shelter

Vocabulary:
Nomads = People who move from place to place to survive.



Why Move?



They had to follow animal migrations and plant cycles. Lived in groups of 20 to 30.

Natural Shelters



Used natural caves for protection from the weather.

Built Shelters



Wood poles or large mammoth bones covered with animal hides (tents).

TEACHER NOTES:
Moving wasn't a choice; it was required to follow the food. Highlight human adaptability: if there was no wood, they built tents out of mammoth bones!

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The Roles of Men and Women



Women's Roles

- Stayed close to camp.
- Bore and raised children.
- Gathered berries, nuts, roots.
- Trapped small animals.
- Kept the camp safe.



Men's Roles

- Traveled far from camp.
- Hunted herds of large animals.

Equality

Both shared the job of finding food. Both made decisions that affected the group. Scientists believe there was equality between them.

TEACHER NOTES:
Emphasize that survival required everyone's hard work. If men couldn't find a large animal, the group survived on the food the women gathered.

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Adapting to Survive: The Discovery of Fire

The Problem

As early hominids (Homo erectus) moved from warm tropics to cold climates, they needed to adapt.



The Solution & Evidence

They learned to use fire. Archaeologists have found piles of ashes in caves proving early humans used fire systematically 500,000 years ago.

TEACHER NOTES:
Define “adapting” (changing habits to survive in a new environment). Stress the incredible timeline: evidence of fire goes back 500,000 years.

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The Power of Fire



Heat: Provided warmth in the freezing cold.



Hunting: Hunters used fire to flush wild pigs out of bushes.





Cooking: Food tasted better and lasted longer.

- Easier to chew and digest.
- Improved human nutrition.



Light: Made it possible to see inside dark caves.



Protection: Kept dangerous wild animals away from the camp.

TEACHER NOTES:
Ask: Before fire, what happened when the sun went down?
Focus on cooking: it changed food chemistry, making it easier to get energy and nutrition.

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How Early Humans Started Fires

Fire-making was invented independently in different parts of the world.

Method 1: Friction



- **Rubbing** pieces of wood together.
- Later developed into sturdy, **drill-like** wooden devices.

Method 2: Sparks



- **Striking** a certain stone (iron pyrites) against a rock.
- The **spark** was used to **light** dry grass or leaves.

TEACHER NOTES:
We know these methods because archaeologists study traditional peoples who use similar techniques today. Friction creates heat; sparks create a flash of burning material.

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Science & Technology: Field Archaeology

How do we learn about early humans? Archaeologists combine old tools with modern technology.



Laptops make the dig easier.

Hand-held GPS (Global Positioning System)

Explorer's Tech Kit



Sketched Field Photo: Archaeologist at work with a brush, revealing a find.

Finding the Site

First, scientists search for promising sites using:

- Computer databases
- Maps
- Aerial photos
- Satellite images
- Hand-held GPS (Global Positioning System)

TEACHER NOTES:

Transition to present day: How do we know this is true if it happened 500,000 years ago?

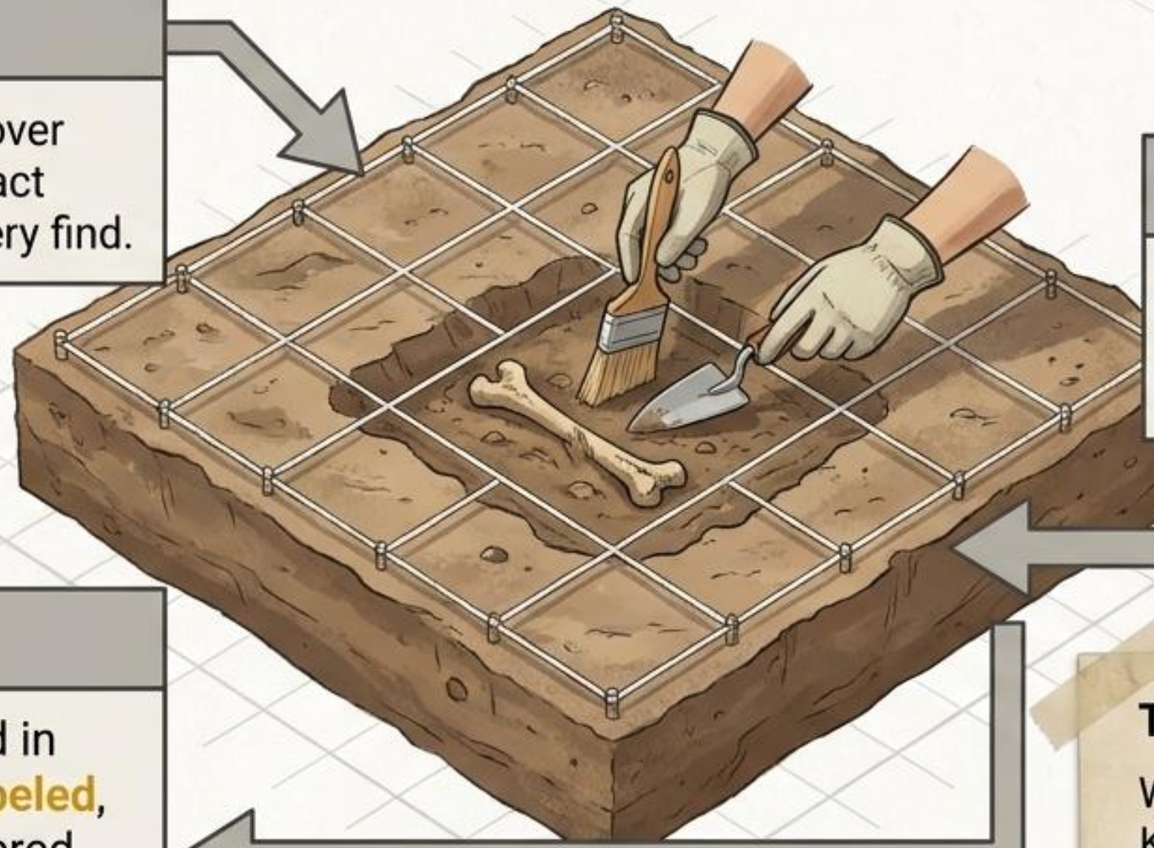
Explain archaeology is like detective work using satellites to find dirt.

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The Archaeological Dig

Step 1: The Grid

The team sets up a grid over the area to record the exact location and depth of every find.



Step 2: Excavation

Workers carefully use **trowels** (small shovels) and **brushes** to unearth artifacts.

Step 3: Recording

Every object is recorded in **field notes**, **cleaned**, **labeled**, **photographed**, and entered into a **computer database**.

TEACHER NOTES:

Why the grid? Context is everything. Knowing exactly how deep an object was helps tell its story and age. Emphasize that digging is incredibly slow, careful work.

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Laboratory Testing: How Old Is It?

After digging up artifacts, scientists use **chemistry** and **biology** in a lab to find the exact date.



Stratigraphy

Dating an object by the soil layer it was found in. Deeper soil is usually older.



Advanced Lab Tests:

- Radiocarbon dating
- Potassium-argon dating
- DNA testing
- Thermoluminescence
- Dendrochronology

TEACHER NOTES:

Do not force students to memorize all the advanced names. Just explain these are scientific tools.

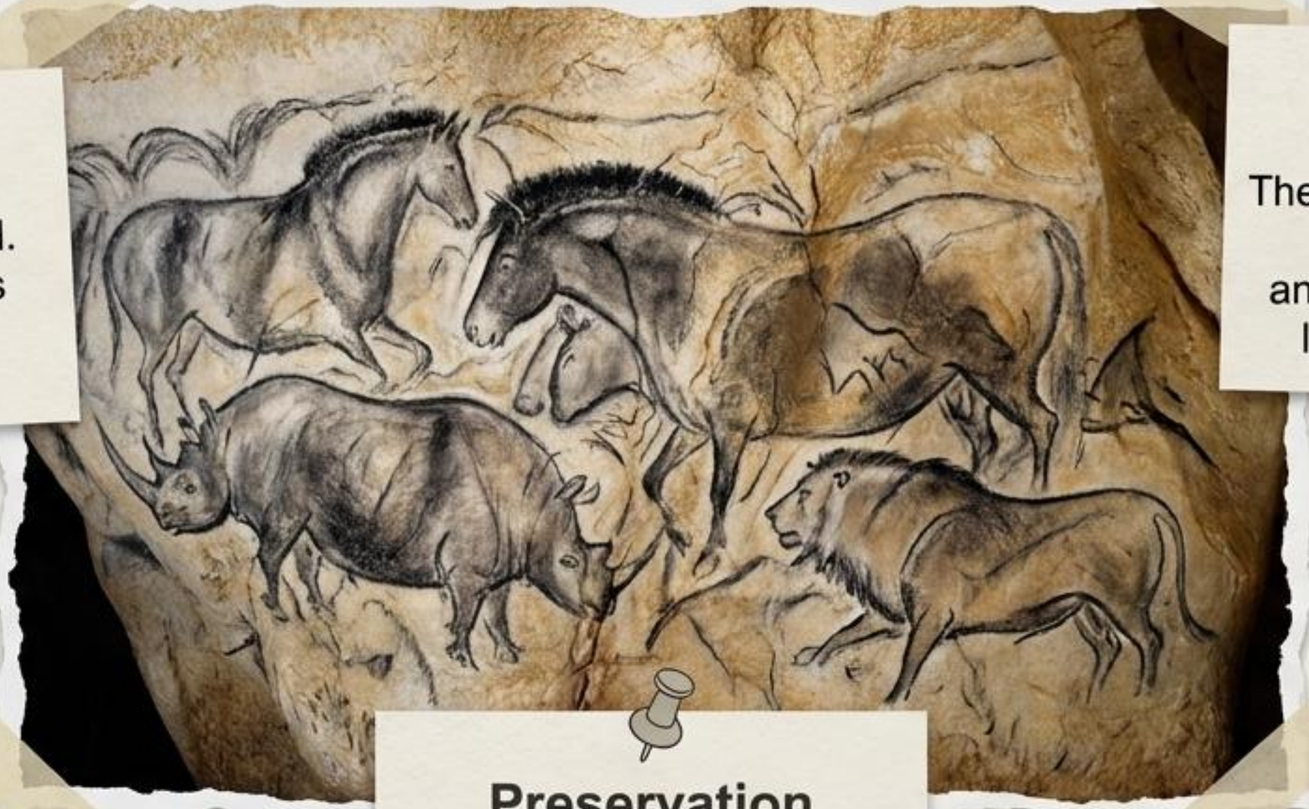
Stratigraphy: deeper equals older.
DNA: traces human relation.

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Primary Source: Chauvet Cave (France)

Age & Materials

Around 35,000 years old. Charcoal from fires was used for drawing.



The Art

The famous “Panel of Horses”. Artists drew dangerous animals (mammoths, rhinos, lions) but rarely humans.

Preservation

Why did it survive? The cave had a constant 56°F temperature and 99% humidity.

TEACHER NOTES:
Point out the beautiful artistry—it looks highly skilled, not primitive.
Ask: Why paint dangerous animals but not themselves?
The climate acted like a museum vault.

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Surviving the Ice Ages (100,000 B.C. to 8000 B.C.)

The Environment

- Sheets of thick ice covered large parts of Europe, Asia, and North America.

The Threat

- Ice Age conditions were a serious threat to human life.

Human Adaptation

- The ability to adapt was crucial.



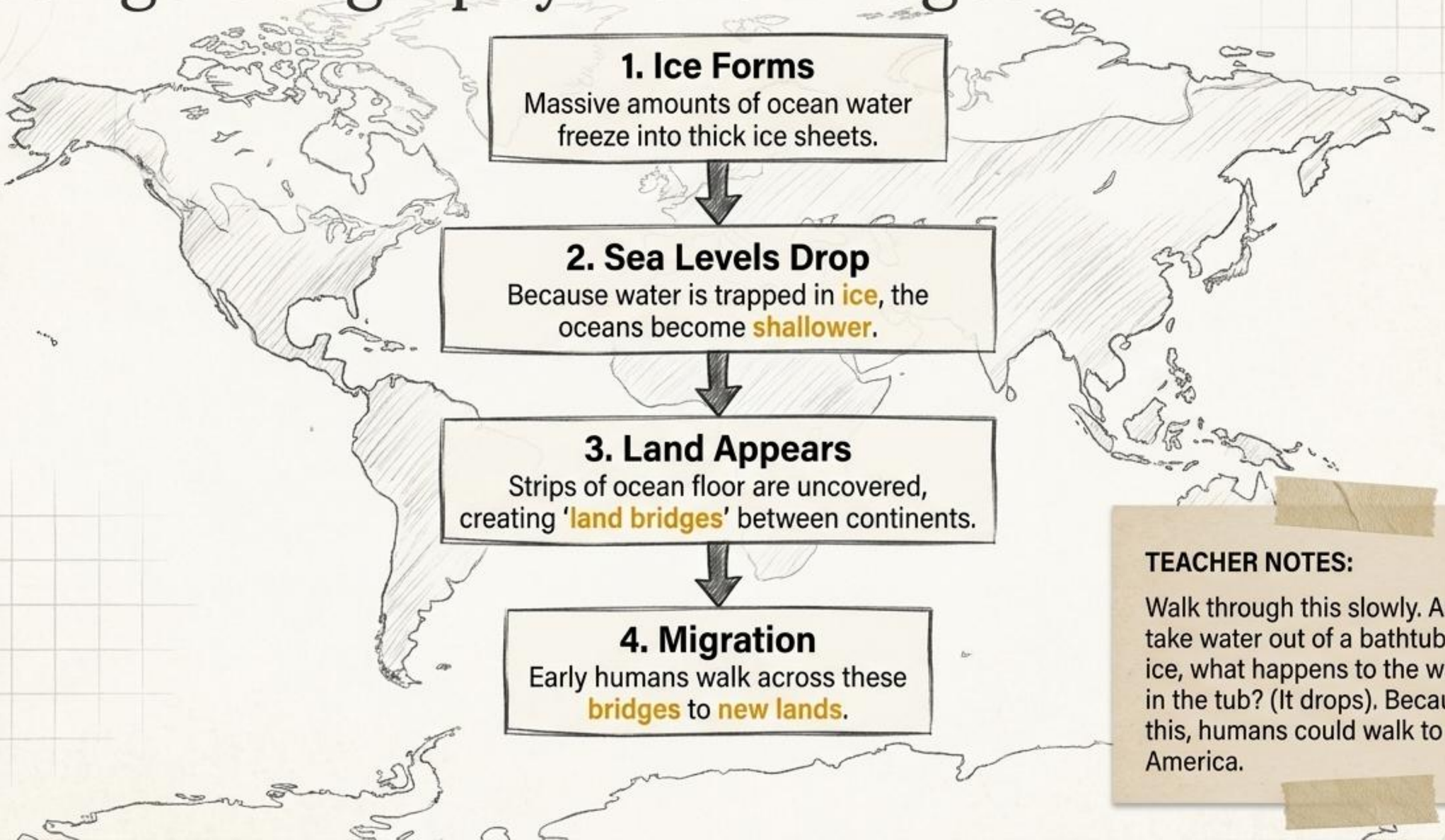
Big Idea: Early humans adapted not by changing their bodies, but by changing their environment (by using fire and building shelters).

TEACHER NOTES:

Explain the Big Idea clearly:
Animals adapt to cold by growing thick fur. Humans adapt by making fire and clothes. We change the environment to fit our needs.

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Ice Age Geography: Land Bridges



1. Ice Forms
Massive amounts of ocean water freeze into thick ice sheets.

2. Sea Levels Drop
Because water is trapped in **ice**, the oceans become **shallower**.

3. Land Appears
Strips of ocean floor are uncovered, creating '**land bridges**' between continents.

4. Migration
Early humans walk across these **bridges** to **new lands**.

TEACHER NOTES:

Walk through this slowly. Ask: If you take water out of a bathtub to make ice, what happens to the water level in the tub? (It drops). Because of this, humans could walk to North America.

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Creating Art: The Great Caves

Altamira (Spain)

- Found: 1879 by a little girl.
- Detail: Paintings were created over a long span of time, not all at once.

Lascaux (France)

- Found: 1940 by children after a bad storm uprooted a tree.
- Detail: Paintings were deteriorating but showed horses, bulls, and stags.

Chauvet (France)

- Found: 1994.
- Detail: Carbon dating showed it was twice as old as the others, yet the drawings were more advanced.

TEACHER NOTES:
Highlight that two major historical sites were discovered by children! Point out the surprise of Chauvet: scientists thought older art would be messy, but it was highly advanced.

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How Cave Art Was Made

Lighting



Stone lamps filled with **animal fat**.

Tools



Fingers and **twigs**.

The Paint



Mineral ores mixed with **animal fat** to make **red**, **yellow**, and **black** colors.

Special Technique



Blowing paint through **hollow reeds** (like a spray can) to make **outlines**.

TEACHER NOTES:
Painting in a pitch-black cave is incredibly difficult. They needed fat lamps just to see. The "hollow reed" technique is the ancient version of spray paint! Mineral ores = natural colored rocks.

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The Meaning of Cave Art: Why Did They Paint?



The Mystery: Humans are drawn as simple stick figures, but animals are painted realistically. Why?

Theory 1: Magic and Religion

Done for religious rituals to bring success in hunting.

Theory 2: Art for Art's Sake

Made simply to please the eye and decorate their living space.

Fact: Art is a basic part of human life, found in prehistory worldwide.

TEACHER NOTES:

Pose this mystery to the class. Discuss Theory 1 vs 2. Explain that archaeologists don't know for sure; they make educated guesses (theories).

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BIG PICTURE: The Blueprint for Human Survival

PALEOLITHIC SURVIVAL



Food

Hunting and gathering in a close relationship with nature.



Technology

Inventing stone hand-axes, spears, and bone tools.



Environment

Creating shelters and using fire for heat, light, and cooking.



Adaptation

Using brains to change the environment during the Ice Age.



Culture

Creating complex art that showed a deep understanding of the world.

TEACHER NOTES:

Use this to summarize the entire lesson. Show how things connect:
better tools = better food.
Fire = surviving the Ice Age.
Smarter brains = art.
Adaptation kept humanity alive.

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Section Review & Critical Thinking

1 Reading Check

Q: What are two important innovations of Paleolithic peoples? (Answer: Tools and Fire)

2 Vocabulary Review

Define: **Paleolithic Age**, **nomad**, **artifact**, **hominid**.

3 Critical Thinking

1. Discuss the difference between the roles of Paleolithic men and women.
2. What do cave paintings tell you about human/animal interactions?
3. Compare and contrast Australopithecus and Neanderthal.

TEACHER NOTES:
Go through questions as a final check. Answers:
Paleolithic = Old stone age.
Nomad = moves to survive.
Roles = men hunted, women gathered, both equal.

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